

NICHOLAS ROBER

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EDUCATION

Massachusetts Institute of Technology

PhD, Aeronautics and Astronautics

Cambridge, MA

2023 – Present

SM, Aeronautics and Astronautics

2023

Thesis: *BReach-LP: a Framework for Backward Reachability Analysis of Neural Feedback Loops*

University of Iowa

BSE, Mechanical Engineering

Iowa City, IA

2020

RESEARCH EXPERIENCE

Massachusetts Institute of Technology

Graduate Research Assistant | Aerospace Controls Lab

Cambridge, MA

2021 – Present

Advisor: Jonathan How

- Conduct industry-sponsored research on verification and synthesis of safe autonomous systems under uncertainty
- Present and defend findings through written journal and conference submissions and presentations at group meetings, conferences, and workshops
- Contribute to writing and conceptualization of funding proposals

University of Iowa

Undergraduate Research Assistant | Cooperative Autonomous Systems Lab

Iowa City, IA

2019 – 2021

Advisor: Venanzio Cichella

- Designed algorithms for motion planning and obstacle avoidance of underwater vehicles
- Compared adaptive and classical control methods and presented findings in a journal publication

AWARDS

Outstanding Student Paper Award

2023

IEEE Aerospace Technical Committee

Backward Reachability Analysis of Neural Feedback Loops

Runner up, Best Paper Award

2022

ICML Workshop for Verification in Machine Learning

Backward Reachability Analysis of Neural Feedback Loops

PUBLICATIONS

Refereed Journal Articles

- mahesh2025safe.
- rober2024constraint.
- rober2023backward.
- rober20223d.
- rober2021three.

Refereed Conference Articles

- rober2024online.
- rober2023hybrid.
- rober2022backward.
- rober2021geometric.

Theses

- rober2023thesis.

TEACHING EXPERIENCE AND TRAINING

Mentorship

MIT

Undergraduate Students

Dylan Gaillard

Fall 2024 - Present

Ryosei Tanakamura

Summer 2025 - Present

Luana Rampelotti

Summer 2025

Guest Lectures

Northeastern University

Verifiable Machine Learning

2023, 2024, 2025

Pedagogical Training

MIT Communications Lab Training Series

2023 - 2024

Undergraduate Teaching Assistantship

The University of Iowa

Control of Mechanical Engineering Systems

Fall 2020

Advanced Linear Control Systems

Spring 2020

Introduction to Engineering Computing

Fall 2018, Fall 2019

Engineering Fundamentals I: Statics

Summer 2018, Summer 2019

PRESENTATIONS

American Control Conference (ACC), Talk

2025

International Conference on Robotics and Automation (ICRA), Talk

2024

Allerton Conference, Invited Talk

2023

American Control Conference (ACC), Talk

2023

Conference on Decision and Control (CDC), Talk

2022

ICML Workshop on Formal Verification of Machine Learning, Talk

2022

ICRA Workshop on Safe and Reliable Robot Autonomy under Uncertainty, Talk

2022

AIAA Scitech Forum, Talk

2021

PROFESSIONAL ACTIVITIES

Community Services

Lead Organizer, Workshop on Formal Verification of Control Systems with NN Components, ACC

2025

Session Co-Chair, Safe Control I, ACC

2025

Student Volunteer, ACC 2025

2025

Institutional Services

Massachusetts Institute of Technology

Fellow, AeroAstro Communications Lab	2023 - Present
Mentor, Graduate Application Assistance Program	2024, 2025
Panelist, MIT AeroAstro Communications Lab Masters Kickoff Seminar	2025
Panelist, MIT Communications Lab Summer Institute	2024
Panelist, MIT Graduate Association of Aeronautics and Astronautics Seminar	2024
Student Liason, LiDS Seminar Speaker Series	2023
Mentor, Freshman Pre-Orientation Program	2022

University of Iowa

Panelist, New Student Seminar	2019
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Review Activities

Journals

Automatica
AIAA Journal of Guidance, Control, and Dynamics (JGCD)
IEEE Control Systems Letters (L-CSS)
IEEE Open Journal of Control Systems (OJ-CSYS)
IEEE Transactions on Automation and Control (TAC)
IEEE Transactions on Robotics (T-RO)
Journal of Field Robotics (JFR)
Nonlinear Analysis: Hybrid Systems
Ocean Engineering
Robotics and Automation Letters (RA-L)

Conferences

American Control Conference (ACC)
Conference on Decision and Control (CDC)
Learning for Dynamics and Control (L4DC)

Workshops

Robotics Science and Systems 2024 Workshop: Towards Safe Autonomy (RSS)